9.Exploring AI-Assisted Features and Built-In Analytics (Zia)

1. Aim

To explore and integrate Zia (Zoho Creator's AI assistance) and built-in analytics into the **Patient Appointment & Scheduling Management** application to improve appointment-form design, assist Deluge scripting, and visualize patient appointment trends for clinic administration.

2. Objective

- Understand how Zia Build Agent and Zia Assistant support natural-language instructions in Zoho Creator.
- Use AI-assisted field suggestions to optimize the Appointment_Booking_Form.
- Explore AI-assisted Deluge validation without duplicating existing business rules.
- Configure graphical analytics showing appointment status and appointments booked per doctor.
- Explore conversational requests for dashboard changes and recognize the limitations of the AI agent.

3. Tools Required

- Zoho Creator account with Zia Build Agent / Zia Assistant access.
- Patient Appointment & Scheduling Management application.
- Appointment_Booking_Form with sample appointment data.
- Clinic_Command_Center_Page for displaying charts and analytics.
- Existing Validate_Appointment form workflow.

4. Procedure

Step 1: Leveraging Zia for Form and Application Optimization

Zia Build Agent interprets natural-language requests and proposes changes to form layouts.

1. Open the Patient Appointment & Scheduling Management application in the Zoho Creator builder.
2. Go to **Design** and open **Appointment_Booking_Form**.
3. In the **Build Agent** panel, prompt Zia to suggest useful appointment-booking fields without duplicating existing fields.
4. Review the proposal for **Consultation Type**, **Visit Reason**, and **Emergency Contact Number**.
5. Click **Apply Changes** and verify the newly added fields in the form builder.

Step 2: Exploring Zia-Assisted Deluge Scripting

Zia Assistant helps generate code suggestions that should be reviewed before being adopted into an existing workflow.

1. Open **Workflow > Form Workflows** and select **Appointment_Booking_Form**.
2. Open the **Validate_Appointment** workflow and its Deluge Script editor.
3. Ask Zia Assistant to generate validation logic preventing appointments with a past preferred date.
4. Compare its suggested code with the existing validation for **Preferred_Appointment_Date**. The workflow already checks whether the date is earlier than **zoho.currentdate**.
5. Keep the existing validation code unchanged to avoid duplicate checks.

Existing date-validation logic:

```
if(input.Preferred_Appointment_Date < zoho.currentdate)
{
  alert "Appointment date cannot be in the past.";
  cancel submit;
}
```

Step 3: Configuring Built-In Appointment Analytics

Create charts from the existing booking records to monitor appointment distribution.

1. Open **Design > Clinic_Command_Center_Page** in the page builder.
2. Add a **Pie Chart** using **Appointment_Booking_Form**, group by **Status**, and use **Count of Total Records**. Name it **Appointment Status Analysis**.
3. Add a **Bar Chart** grouped by **Doctor Name**, using **Count of Total Records**. Name it **Appointments Booked Per Doctor**.
4. Enable Data Labels, save the configurations, and preview the page in the live application.

Step 4: Interacting with Zia Build Agent for Analytics

1. Open the **Build Agent** panel while editing **Clinic_Command_Center_Page**.
2. Enter the prompt: *"Show the number of appointments booked for each doctor in a bar chart using the existing Appointment_Booking_Form."*
3. Review Zia's response. The agent reported that it could not automatically add the chart because it could not retrieve the required complete page structure.
4. Create the doctor-wise chart manually through the Zoho Creator chart configuration interface, then verify it in the running clinic dashboard.
5. Review the analytics shown in the dashboard, including total appointments, status-wise breakdown, department-wise distribution, and doctor-wise counts.

Note: Zia provided a successful form change and a Deluge suggestion, but the doctor-wise bar chart was added through the manual chart builder after the agent reported a limitation.

5. Screenshot

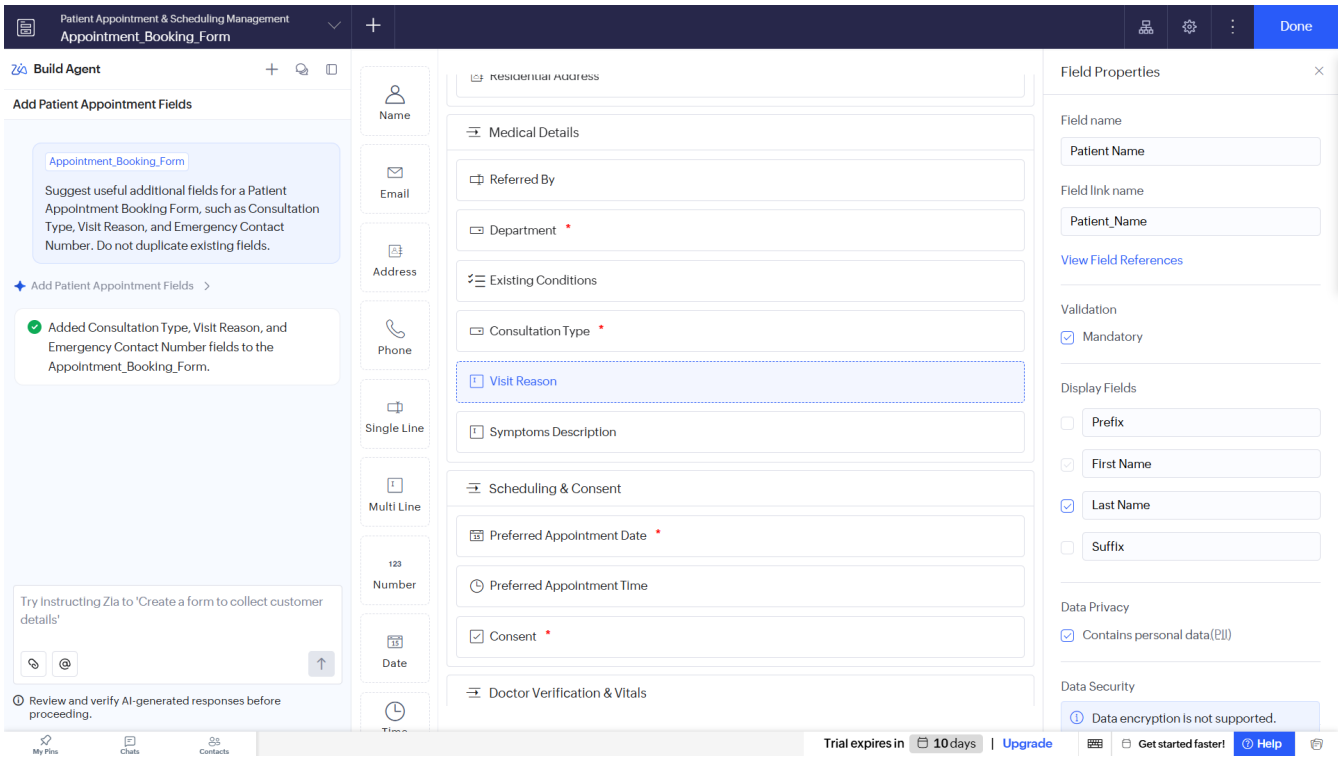


Figure 1: Zia Build Agent confirms the addition of three fields to Appointment_Booking_Form.

The AI-assisted form update added Consultation Type, Visit Reason, and Emergency Contact Number to the existing patient booking form.



Figure 2: Zia Assistant suggests Deluge appointment-date validation; the existing workflow contains the same check.

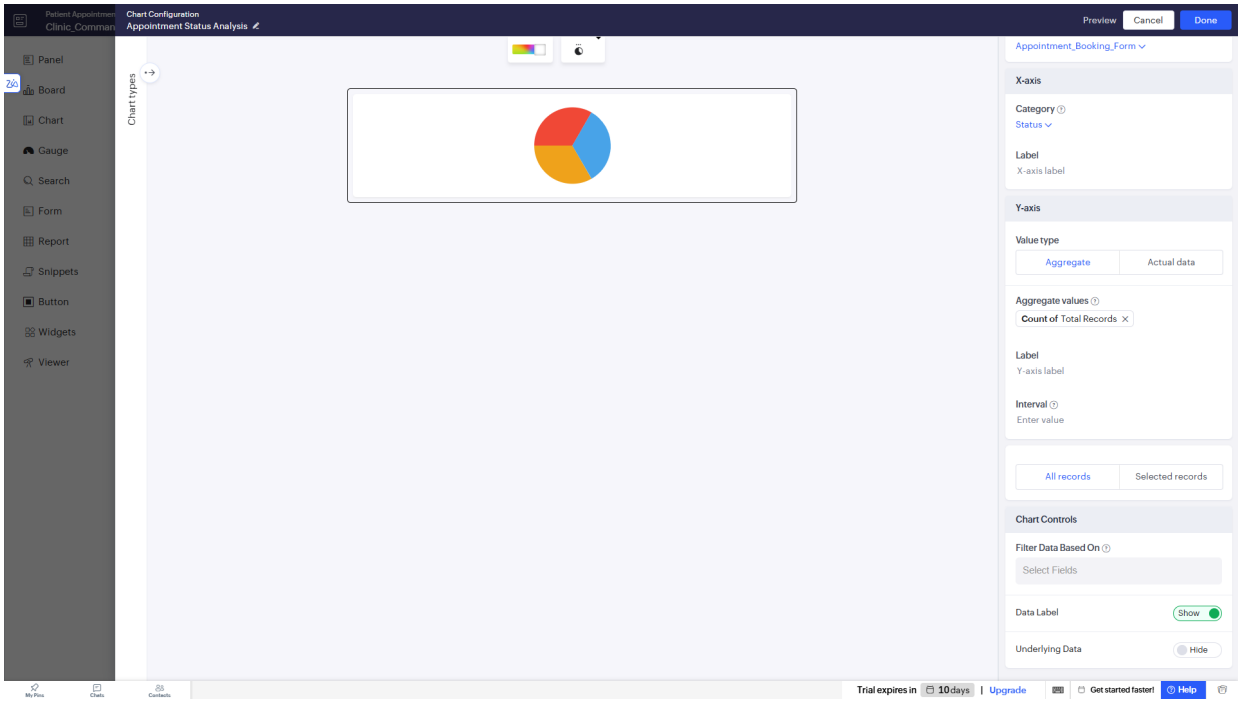


Figure 3: Pie chart configuration for appointment status using Status and record count.

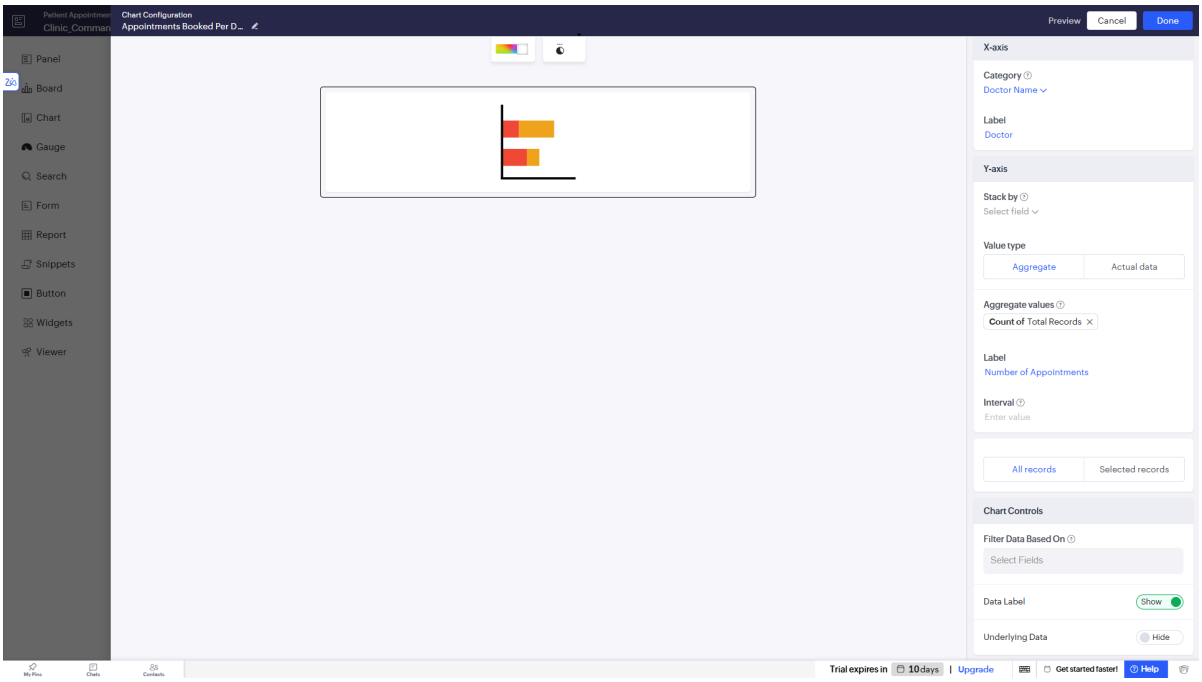


Figure 4: Doctor-wise bar chart configuration, grouped by Doctor Name with total record count.

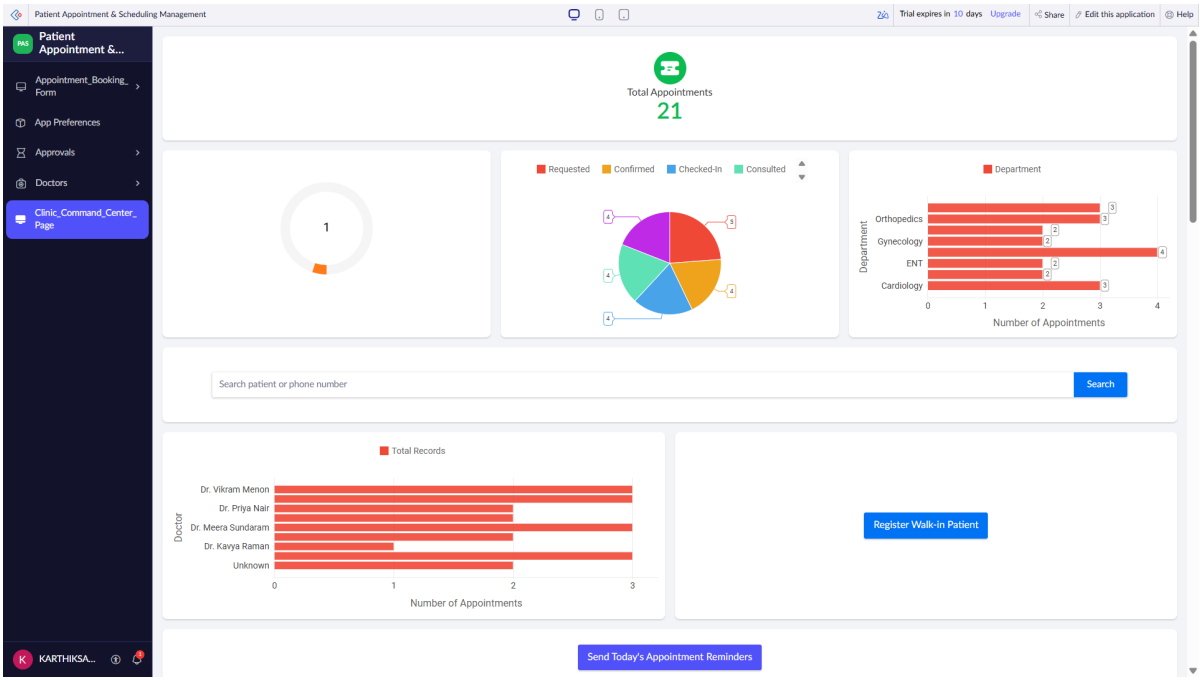


Figure 5: Live Clinic Command Center dashboard displaying appointment, department, and doctor analytics.

6. Result

AI-assisted features and built-in analytics were explored and applied in the **Patient Appointment & Scheduling Management** application using Zoho Creator. Zia Build Agent successfully added Consultation Type, Visit Reason, and Emergency Contact Number to the Appointment_Booking_Form. Zia Assistant generated a Deluge validation suggestion for preventing past-date bookings; the existing equivalent validation was retained to avoid duplication.

Built-in analytics charts were configured for appointment status and doctor-wise appointment counts, and the Clinic Command Center displayed appointment statistics from the application data. When Zia Build Agent could not automatically add a doctor-wise chart, the chart was configured manually. The experiment demonstrated the benefits and practical limitations of AI-assisted development together with visual, data-driven clinic analytics.